

Coordinate System & Symmetry Specification for Warp-Bubble Metrics

Arcticoder

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1 Coordinate Chart

We adopt a standard spherical coordinate system on the background spacetime:

Coordinates : (t, r, θ, ϕ)

Domains : $t \in (-\infty, +\infty), \quad r \in [0, +\infty), \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi).$

2 Symmetry Assumptions

1. **Axial symmetry** about the z -axis: no ϕ -dependence.
2. **Equatorial reflection symmetry**: invariance under $\theta \mapsto \pi - \theta$.
3. **Compact support in r** : warp profiles A, B, C, D nontrivial only for $r \leq R$.

3 Reduced Metric Ansatz

Imposing the above, the line element becomes

$$ds^2 = -A(r) dt^2 + B(r) dr^2 + C(r) r^2 d\theta^2 + D(r) r^2 \sin^2 \theta d\phi^2,$$

with

$$A(r), B(r), C(r), D(r) = \begin{cases} 1, & r > R, \\ \text{warp-bubble profiles}, & r \leq R, \end{cases}$$

and boundary conditions $A(R) = B(R) = C(R) = D(R) = 1$.

4 Worked Example (Sketch)

- Eliminate cross-terms by staticity & symmetry.
- Enforce θ -independence $\Rightarrow$ functions of r only.
- Impose $g_{\mu\nu} = \eta_{\mu\nu}$ for $r > R$.
- Rename $g_{tt} = -A(r)$, $g_{rr} = B(r)$, etc.

5 Automation Scripts

To generate, test and document metric ansätze for any warp-bubble profile, include a ‘scripts/’ folder with:

```
warp-bubble-coordinate-spec/  
|-- coordinate_spec.tex  
|-- scripts/  
|   |-- fetch_shape.py  
|   |-- generate_ansatz.py  
|   +-- requirements.txt  
+-- metrics/  
    +-- <generated .tex snippets>
```

1. `fetch_shape.py`

- *Purpose:* download shape-profile data from <https://arcticoder.github.io/warp-bubble-shape-catalog> and convert to JSON.
- *Data Sources:*
 - Primary: NPZ format (higher precision) from <https://github.com/arcticoder/warp-bubble-shape-catalog/raw/refs/heads/main/data/profiles>
 - Fallback: CSV format from <https://raw.githubusercontent.com/arcticoder/warp-bubble-shape-catalog/refs/heads/main/data/profiles>
- *Usage:*

```
python scripts/fetch_shape.py --shape alcubierre  
python scripts/fetch_shape.py --shape natario --format csv
```

- *Behavior:*
 1. Perform HTTP GET on the NPZ data (or CSV if specified).
 2. Parse the data and extract the requested shape profile along with r values.
 3. Convert to JSON format with metadata and save to `scripts/shapes/{shape}.json`.

2. `generate_ansatz.py`

- *Purpose:* produce a LaTeX snippet defining $A(r), B(r), C(r), D(r)$ from the fetched shape.
- *Usage:*

```
python scripts/generate_ansatz.py \  
    --shape alcubierre \  
    --template ../coordinate_spec.tex \  
    --out metrics/alcubierre_ansatz.tex
```

- *Behavior:*

1. Load `scripts/shapes/{shape}.json`.
2. Use Jinja2 (or simple string templates) to substitute:

$$A(r) = 1 - f(r), \quad B(r) = \frac{1}{1 - f(r)}, \dots$$

3. Emit a standalone TeX fragment in `metrics/`.

Dependencies & Installation

Add a `scripts/requirements.txt`:

```
requests>=2.25
jinja2>=3.0
sympy>=1.8      # if you extend to symbolic curvature computation
numpy>=1.20     # for handling NPZ data format
```

Install with:

```
pip install -r scripts/requirements.txt
```

Each script implicitly references the "shape catalog" at <https://arcticoder.github.io/warp-bubble-shape-catalog/> as its single source-of-truth. The data is provided in both NPZ format (higher precision, 500 data points) and CSV format for compatibility.

Available shapes include: **alcubierre**, **natario**

Data sources:

- NPZ: <https://github.com/arcticoder/warp-bubble-shape-catalog/raw/refs/heads/main/data/profiles/alcubierre.npz>
- CSV: <https://raw.githubusercontent.com/arcticoder/warp-bubble-shape-catalog/refs/heads/main/data/profiles/alcubierre.csv>

For offline work, you may instead add that repo as a git submodule:

```
git submodule add \
  https://github.com/arcticoder/warp-bubble-shape-catalog \
  scripts/warp-bubble-shape-catalog
```

6 Testing & Validation

To verify the automation pipeline works correctly:

1. Test data fetching:

```
python scripts/fetch_shape.py --shape alcubierre
```

This should create `scripts/shapes/alcubierre.json` with shape profile data.

2. Test ansatz generation:

```
python scripts/generate_ansatz.py \
--shape alcubierre \
--out metrics/alcubierre_test.tex
```

This should create a LaTeX snippet defining the metric components.

3. Verify the output contains proper mathematical expressions:

$$A(r) = 1 - (\text{interpolated_alcubierre}(r)), \quad B(r) = 1/(1 - (\text{interpolated_alcubierre}(r)))$$

The scripts automatically handle data format conversion, error checking, and fallback between NPZ and CSV sources. All generated content includes proper LaTeX formatting for mathematical expressions.

7 Documentation & Deployment

This specification is available as a live website with MathJax rendering at:

<https://arcticoder.github.io/warp-bubble-coordinate-spec/>

The documentation is automatically built and deployed using GitHub Actions whenever changes are pushed to the main branch. The Jekyll site includes:

- Mathematical equation rendering via MathJax
- Responsive layout for desktop and mobile
- Automatic syntax highlighting for code blocks
- Integration with the warp-bubble-shape-catalog data sources

The automation pipeline ensures that both the LaTeX document and website remain synchronized with the latest shape data and computational tools.

References

References

- [1] M. Alcubierre, *The Warp Drive: Hyper-fast travel within general relativity*, Classical and Quantum Gravity **11** (1994) L73–L77.
- [2] J. Natário, *Warp drive with zero expansion*, Classical and Quantum Gravity **19** (2002) 1157–1165.
- [3] Arcticoder, *Warp Bubble Shape Catalog*, GitHub Pages, 2025.
<https://arcticoder.github.io/warp-bubble-shape-catalog/>
- [4] Arcticoder, *Warp Bubble Coordinate Spec*, GitHub Pages, 2025.
<https://arcticoder.github.io/warp-bubble-coordinate-spec/>